THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME III

The Ferrohydrodynamic System Assembled,
Elimination of the Particle Spin,
and the Alternating Field: Equations (SM-6)–(SM-15)

Where Volume II ended. Volume II built the statistical mechanics of the orientations (Boltzmann law, Langevin function $L(\xi) = \coth\xi - 1/\xi$, equilibrium magnetisation $\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$), rotational Brownian motion (the Smoluchowski equation, the Einstein relation $D_r = kT/\zeta_r$, and $\tau_B = 3\eta V/kT$), the magnetisation relaxation equation, and the bridge from magnetic torque to viscosity, $\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$. With those it derived Eqs. (SM-2), (SM-3), (SM-4) and (SM-5).

Volume III now presents, in the papers' own order and in the eight-part format, the complete system of ferrohydrodynamic equations (SM-6)–(SM-10), the elimination of the particle spin (SM-11)–(SM-13), and the specification of the alternating field (SM-14)–(SM-15). Several of these were derived in Volume II as tools; where that is so, the derivation is *cited*, not repeated, and the chapter concentrates on the context, the assumptions, the physical reading and the numerical checks. Two genuinely new derivations appear here: the Kelvin force $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ in Chapter 19, and the magnetostatic constraints (SM-9) in Chapter 22. Chapter numbers resume at 19, page numbers at 78.

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Part VII

The Ferrohydrodynamic System, Assembled

Chapter 19

Equation (SM-6): The Fluid-Motion Equation

The equation is

$$\rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M} \cdot \nabla)\boldsymbol{H} + \eta \Delta \boldsymbol{v} + \frac{I}{2\tau_s} \nabla \times \big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{SM-6}$$

19.1 Context

Shliomis & Morozov open Sec. II by stating that "the system of ferrohydrodynamic equations accounting for the connection between the magnetic and rotational mechanical degrees of freedom consists of the fluid motion equation, the magnetization equation, and the equation of rotational motion of the particles". Equation (SM-6) is the first of these three. It is the ordinary Navier–Stokes equation of Sec. 5.4 with two additional terms: one describing the force a non-uniform field exerts on a magnetised medium, and one describing the momentum transferred to the liquid when the particles spin at a rate different from the local vorticity. The second is the term that carries the entire magnetoviscous effect; the first plays no role in the papers' calculations (the applied field is uniform) but must be present for the equation to be correct in general, and we derive it.

19.2 Symbols

Symbol	Meaning	Type	Units
ρ	mass density of the ferrofluid as a whole (carrier plus grains)	scalar	$\mathrm{g\,cm^{-3}}$
$\boldsymbol{v}$	velocity field of the ferrofluid, treated as a single continuum	vector field	$\mathrm{cm\,s^{-1}}$
$\mathrm{d}/\mathrm{d}t$	material derivative $\partial_t + (\boldsymbol{v}\cdot\nabla)$, Eq. (3.2.3)	operator	$\mathrm{s^{-1}}$
p	pressure	scalar field	$\mathrm{dyn\,cm^{-2}}$
$\boldsymbol{M}$	magnetisation	vector field	G
$\boldsymbol{H}$	magnetic field acting inside the fluid	vector field	Oe
η	viscosity of the suspension in zero field	scalar	P
Δ	the authors' notation for the Laplacian ∇^2	operator	$\mathrm{cm^{-2}}$
I	moment of inertia of the particles per unit volume, $I = nI_p$	scalar	$\mathrm{g\,cm^{-1}}$
τ_s	inertial spin relaxation time, Eq. (SM-1)	scalar	s
$\boldsymbol{\Theta}$	mean angular velocity of the particles	vector field	$\mathrm{s^{-1}}$
$\boldsymbol{\Omega}$	local angular velocity of the fluid, $\frac{1}{2}\nabla\times\boldsymbol{v}$	vector field	$\mathrm{s^{-1}}$

Remark 19.2.1 (One velocity field for two phases). Writing a single $\boldsymbol{v}$ for a suspension of solid grains in a liquid is an approximation whose justification is the enormous separation of scales: the grains are 10 nm, the flow varies over 1 mm, a ratio of 10^5. On any scale the flow can resolve, grain and liquid move together. What the grains *do* retain as an independent degree of freedom is their *spin* $\boldsymbol{\Theta}$, because a torque can make them rotate differently from the fluid without making them translate differently. This is the entire physical content of ferrohydrodynamics, and it is why $\boldsymbol{\Theta}$ appears in (SM-6) while no second velocity does.

19.3 Assumptions

1. Incompressible ($\nabla\cdot\boldsymbol{v} = 0$) and Newtonian carrier, so that the ordinary viscous term is $\eta\nabla^2\boldsymbol{v}$ (Sec. 5.4).

2. The suspension is treated as a single continuum with one velocity field, as in the remark above.

3. The magnetic grains are rigid dipoles, so a magnetic torque on the moment is a mechanical torque on the grain (Sec. 17.6).

4. Gravity and buoyancy are omitted: they are balanced in the horizontal capillary of the experiment and drop out of the equation for the deviation from hydrostatic equilibrium.

5. The intrinsic spin angular momentum of the grains is negligible compared with their orbital angular momentum, as established by the smallness of τ_s (Eq. SM-1).

19.4 Mathematical Background: the Kelvin Force

Result 19.4.1 (Gradient of a scalar product, constant first factor)**.** *For a* constant *vector* $\boldsymbol{A}$ *and a vector field* $\boldsymbol{B}(\boldsymbol{r})$*,*

$$\nabla\big(\boldsymbol{A}\cdot\boldsymbol{B}\big) = (\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + \boldsymbol{A}\times(\nabla\times\boldsymbol{B}). \tag{19.4.1}$$

Derivation. Work in components. The left-hand side is

$$\Big[\nabla(\boldsymbol{A}\cdot\boldsymbol{B})\Big]_i = \partial_i\big(A_jB_j\big) = A_j\,\partial_iB_j, \tag{19.4.2}$$

the derivative passing over A_j because $\boldsymbol{A}$ is constant. For the right-hand side, the first term is immediately

$$\big[(\boldsymbol{A}\cdot\nabla)\boldsymbol{B}\big]_i = A_j\partial_jB_i. \tag{19.4.3}$$

The second term is evaluated with the ε–δ identity (1.6.1):

$$\begin{aligned}\big[\boldsymbol{A}\times(\nabla\times\boldsymbol{B})\big]_i &= \varepsilon_{ijk}A_j\big(\nabla\times\boldsymbol{B}\big)_k && (19.4.4)\\ &= \varepsilon_{ijk}A_j\,\varepsilon_{klm}\partial_lB_m && (19.4.5)\\ &= \varepsilon_{ijk}\varepsilon_{klm}\,A_j\partial_lB_m && (19.4.6)\\ &= \big(\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}\big)A_j\partial_lB_m && (19.4.7)\\ &= A_j\partial_iB_j - A_j\partial_jB_i. && (19.4.8)\end{aligned}$$

Adding (19.4.3) and (19.4.8), the terms $A_j\partial_jB_i$ cancel and what remains is $A_j\partial_iB_j$, which is (19.4.2). ■

Result 19.4.2 (Kelvin force density)**.**

$$\boldsymbol{f}_{mag} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}. \tag{19.4.9}$$

Derivation, in four steps. *Step 1.* A single rigid dipole of moment $\boldsymbol{m}$ at position $\boldsymbol{r}$ has the potential energy $U(\boldsymbol{r}) = -\boldsymbol{m}\cdot\boldsymbol{H}(\boldsymbol{r})$ of Eq. (4.4.2). Its *orientation* is fixed at the instant considered, so the only $\boldsymbol{r}$ dependence is through the field.

Step 2. The force on a body is minus the gradient of its potential energy (this is the definition of a conservative force; it follows from requiring that the work done, $\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{r}$, equal $-\mathrm{d}U$ for every displacement):

$$\boldsymbol{F} = -\nabla U = \nabla\big(\boldsymbol{m}\cdot\boldsymbol{H}\big). \tag{19.4.10}$$

Step 3. Apply (19.4.1) with $\boldsymbol{A}=\boldsymbol{m}$ (constant) and $\boldsymbol{B}=\boldsymbol{H}$:

$$\boldsymbol{F} = (\boldsymbol{m}\cdot\nabla)\boldsymbol{H} + \boldsymbol{m}\times\big(\nabla\times\boldsymbol{H}\big). \tag{19.4.11}$$

The second term vanishes because $\nabla\times\boldsymbol{H}=0$ in the non-conducting, quasi-static situation — this is the second relation of Eq. (SM-9), derived independently in Chapter 22. Hence $\boldsymbol{F} = (\boldsymbol{m}\cdot\nabla)\boldsymbol{H}$.

Step 4. Sum over the n grains in unit volume. The field and its derivatives are the same for all of them (they vary over millimetres, the grains are nanometres apart), so the sum acts only on the moments:

$$\boldsymbol{f}_{\mathrm{mag}} = n\big\langle(\boldsymbol{m}\cdot\nabla)\big\rangle\boldsymbol{H} = \Big(\big(n\langle\boldsymbol{m}\rangle\big)\cdot\nabla\Big)\boldsymbol{H} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}, \tag{19.4.12}$$

using $\boldsymbol{M} = n\langle\boldsymbol{m}\rangle$ from Eq. (4.5.1). ■

Dimensional check. $[\boldsymbol{M}][\boldsymbol{H}]/\mathrm{cm} = \mathrm{G{\cdot}Oe/cm}$. In Gaussian units G·Oe is an energy density, $\mathrm{erg\,cm^{-3}}$, so the quotient is $\mathrm{erg\,cm^{-4}} = \mathrm{dyn\,cm^{-3}}$: a force per unit volume. ✓

19.5 Step-by-Step Derivation of (SM-6)

Step 1: start from Cauchy's equation. From Eq. (5.2.4),

$$\rho \frac{\mathrm{d}v_i}{\mathrm{d}t} = f_i + \partial_j \sigma_{ij}, \tag{19.5.1}$$

where $\boldsymbol{f}$ collects all body forces and σ_{ij} is the stress.

Step 2: identify the body force. The only body force is magnetic, so $\boldsymbol{f} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ by (19.4.9).

Step 3: split the stress. As established in Sec. 14.3, the stress of a ferrofluid has a symmetric Newtonian part and an antisymmetric part produced by the body couple exerted on the liquid by the spinning grains:

$$\sigma_{ij} = \underbrace{-p\delta_{ij} + 2\eta e_{ij}}_{\text{Newtonian}} + \underbrace{\tfrac{1}{2}\varepsilon_{ijl}\Lambda_l}_{\text{antisymmetric}}, \qquad \boldsymbol{\Lambda} = \text{body-couple density}. \tag{19.5.2}$$

Taking the divergence, the Newtonian part gives $-\nabla p + \eta\nabla^2\boldsymbol{v}$ exactly as in Sec. 5.4, and the antisymmetric part gives $\frac{1}{2}\nabla\times\boldsymbol{\Lambda}$ by Eq. (14.3.1).

Step 4: identify the couple density. The couple delivered to the liquid is the viscous torque the grains exert on it, which is minus the torque the liquid exerts on the grains. From Eq. (6.5.8) summed over unit volume, the liquid exerts $-n\zeta_r(\boldsymbol{\Theta}-\boldsymbol{\Omega})$ on the grains, so the grains exert

$$\boldsymbol{\Lambda} = +n\zeta_r\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) = \frac{I}{\tau_s}\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) \tag{19.5.3}$$

on the liquid, where the identity $n\zeta_r = I/\tau_s = 6\eta\varphi$ of Eq. (7.5.12) was used. Note the direction: if the particles spin *faster* than the fluid ($\boldsymbol{\Theta} > \boldsymbol{\Omega}$) they hand angular momentum *to* the fluid. That single sign is the germ of the negative-viscosity effect.

Step 5: assemble. Substituting Steps 2–4 into (19.5.1):

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\nabla^2\boldsymbol{v} + \frac{I}{2\tau_s}\nabla\times\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big), \tag{SM-6}$$

which is Eq. (SM-6) with every term derived. ■

19.6 Physical Interpretation

Read term by term, from left to right:

- $\rho\,\mathrm{d}\boldsymbol{v}/\mathrm{d}t$: the inertia of a fluid parcel. Under the conditions of these experiments it is small (Sec. 5.5) but it is kept because (SM-6) is meant to be a general equation.

- $-\nabla p$: the ordinary pressure gradient. In the experiment it is what drives the flow, being supplied by the difference h in ferrofluid level between the ends of the capillary.
- $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$: the Kelvin force. A magnetised medium is pulled towards regions of stronger field. It is what makes ferrofluids climb magnets and form the famous spikes; it plays *no* role in these two papers because the solenoid field is uniform over the capillary, so $\nabla\boldsymbol{H}\approx 0$.
- $\eta\nabla^2\boldsymbol{v}$: viscous diffusion of momentum.
- $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$: the new term. It is nonzero *only* when the grains spin at a rate different from the fluid, which happens *only* when a magnetic torque acts on them. It is a genuine internal source of linear momentum for the fluid, fed by the field.

What if the last term were removed? The equation would reduce to ordinary Navier–Stokes, the viscosity would be the field-free η, and neither the McTague effect nor the negative-viscosity effect would exist. Every result of both papers lives in that one term.

19.7 Connection to the Next Equation

Equation (SM-6) contains three unknown fields: $\boldsymbol{v}$, $\boldsymbol{M}$ and $\boldsymbol{\Theta}$ (the pressure being fixed by incompressibility). It cannot be solved alone. The magnetisation obeys its own dynamics, which is the next equation, (SM-7).

Chapter 20

Equation (SM-7): The Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{SM-7}$$

20.1 Context

The second member of the ferrohydrodynamic triple. It says how the magnetisation of the suspension changes: it is carried around by the rotating particles, and it relaxes towards its instantaneous equilibrium value with the Brownian time τ_B. It is the equation that introduces *memory* into the problem, and without memory — that is, if τ_B were zero — there would be no negative-viscosity effect at all, because $\boldsymbol{M}$ would always be parallel to $\boldsymbol{H}$ and the torque $\boldsymbol{M} \times \boldsymbol{H}$ would vanish identically.

20.2 Symbols

All have appeared: $\boldsymbol{M}$ (magnetisation, G), $\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$ (instantaneous-equilibrium magnetisation, Eq. (10.4.1), G), $\boldsymbol{\Theta}$ (mean particle angular velocity, s^{-1}), $\tau_B = 3\eta V/kT$ (Brownian relaxation time, Eq. (12.3.4), s). The derivative $\mathrm{d}/\mathrm{d}t$ is again the material derivative (3.2.3), since $\boldsymbol{M}$ is a property carried by the moving fluid.

20.3 Assumptions

1. Rigid dipoles (no Néel relaxation): checked numerically in Sec. 17.6.

2. Dilute in the magnetic sense: no dipole–dipole interaction between grains, so that each moment feels only the applied field.

3. Monodisperse grains: a single m and a single τ_B. Real samples are polydisperse, which smears the relaxation over a range of times; this is one of the reasons the measured curves are smoother than the theoretical ones.

4. **The closure assumption.** As shown in Sec. 13.2, the exact first-moment equation contains $\langle e_i e_j \rangle$, and (SM-7) follows only if that second moment is replaced by its isotropic

value $\frac{1}{3}\delta_{ij}$. The equation is therefore *exact* for $\xi \ll 1$ and a controlled approximation otherwise.

20.4 Derivation

Carried out in full in Sec. 13.2, by multiplying the Smoluchowski equation (11.6.1) by $\boldsymbol{e}$ and integrating over orientations. The three contributions were

$$\text{flow:} \quad \boldsymbol{\Omega} \times \langle \boldsymbol{e} \rangle \qquad \text{Eq. (13.2.4)}, \tag{20.4.1}$$

$$\text{diffusion:} \quad -2D_r \langle \boldsymbol{e} \rangle \qquad \text{Eq. (13.2.6)}, \tag{20.4.2}$$

$$\text{magnetic:} \quad +2D_r \langle \boldsymbol{e} \rangle_{\text{eq}} \qquad \text{Eq. (13.2.10)}, \tag{20.4.3}$$

whose sum, multiplied by nm and written with $2D_r = 1/\tau_B$, is Eq. (13.2.12):

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{20.4.4}$$

Section 13.3 then showed that replacing $\boldsymbol{\Omega}$ by the *total* mean particle spin $\boldsymbol{\Theta}$ merely re-books the magnetic torque term, so that (20.4.4) and (SM-7) are the same statement. Nothing is repeated here.

20.5 Physical Interpretation

The convective term $\boldsymbol{\Theta} \times \boldsymbol{M}$. If the grains rotate rigidly with angular velocity $\boldsymbol{\Theta}$, the moments frozen in them rotate too, and a vector rotating with angular velocity $\boldsymbol{\Theta}$ obeys $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta} \times \boldsymbol{M}$ (Sec. 3.4, Step 1). This term conserves $|\boldsymbol{M}|$: indeed $\boldsymbol{M} \cdot (\boldsymbol{\Theta} \times \boldsymbol{M}) = 0$ by (1.5.7), so it can only turn the magnetisation, never change its length.

The relaxation term. Everything that *does* change $|\boldsymbol{M}|$ is in $-(\boldsymbol{M} - \boldsymbol{M}_0)/\tau_B$: thermal agitation degrades any alignment not sustained by the field, and the field rebuilds it, the balance being $\boldsymbol{M}_0$. The single time constant τ_B is the closure's simplification of what is in reality a spectrum.

Why memory matters. Set $\tau_B \to 0$: the relaxation term becomes infinitely stiff, forcing $\boldsymbol{M} = \boldsymbol{M}_0 \parallel \boldsymbol{H}$ at every instant. Then $\boldsymbol{M} \times \boldsymbol{H} = 0$, the coupling term in (SM-6) vanishes, $\boldsymbol{\Theta} = \boldsymbol{\Omega}$, and the ferrofluid is an ordinary liquid. *The entire subject exists because $\tau_B \neq 0$.*

20.6 Connection to the Next Equation

Equation (SM-7) contains $\boldsymbol{\Theta}$, which is still undetermined. The third member of the system supplies it: Newton's law for the rotation of the particles, (SM-8).

Chapter 21

Equation (SM-8): Rotational Motion of the Particles

$$I\,\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M}\times\boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big). \tag{SM-8}$$

21.1 Context

The third equation of the system. It is Newton's second law for rotation [Eq. (4.2.2)] applied to the grains contained in unit volume, with the two torques that act on them: magnetic and viscous. Volume I derived its field-free version as Eq. (7.5.3); here the magnetic torque is restored and the equation is written per unit volume, exactly as the paper does.

21.2 Symbols

I is the sum of the particles' moments of inertia over unit volume, $I = nI_p = n\pi\rho_s d^5/60$ (units $\mathrm{g\,cm^{-1}}$); all other symbols are as before. Both sides have the units of a torque density, $\mathrm{erg\,cm^{-3}}$: $[I][\Theta]/[t] = \mathrm{g\,cm^{-1}s^{-2}} = \mathrm{dyn\,cm^{-2}} = \mathrm{erg\,cm^{-3}}$, and $[M][H] = \mathrm{G\,Oe} = \mathrm{erg\,cm^{-3}}$. ✓

21.3 Assumptions

1. Spherical grains, so that the moment of inertia is a scalar and $\boldsymbol{L} = I\boldsymbol{\Theta}$ (Sec. 4.3).

2. Stokes friction for the viscous torque (Chapter 6).

3. Rigid dipoles, so the magnetic torque on the moment is transmitted to the grain.

4. **The random Brownian torques average to zero.** This deserves emphasis. Each individual grain is battered by fluctuating molecular torques far larger than the systematic ones. But $\boldsymbol{\Theta}$ is defined as an average over the many grains in a physically small volume element (Sec. 4.5), and the random torques, having zero mean and being uncorrelated between grains, cancel in that average. Their effect does not disappear from the theory — it reappears, in the only place it can, as the relaxation term of (SM-7). *The systematic equation (SM-8) and the statistical equation (SM-7) are the two faces of the same molecular bombardment.*

21.4 Step-by-Step Derivation

Step 1. For a single grain, Eq. (4.2.2) with $\boldsymbol{L} = I_p \boldsymbol{\theta}$:

$$I_p \frac{\mathrm{d}\boldsymbol{\theta}}{\mathrm{d}t} = \boldsymbol{T}_{\mathrm{mag}} + \boldsymbol{T}_{\mathrm{visc}} = \boldsymbol{m} \times \boldsymbol{H} - \zeta_r \big(\boldsymbol{\theta} - \boldsymbol{\Omega}\big), \tag{21.4.1}$$

the magnetic torque from Eq. (4.4.2) and the viscous one from Eq. (6.5.8), the latter written in the frame co-rotating with the fluid as explained in Sec. 7.5, Step 2.

Step 2. Sum (21.4.1) over the n grains in unit volume and divide by nothing (we *want* the per-unit-volume form). Averaging termwise:

$$n I_p \frac{\mathrm{d}\langle\boldsymbol{\theta}\rangle}{\mathrm{d}t} = n\langle\boldsymbol{m}\rangle \times \boldsymbol{H} - n\zeta_r \big(\langle\boldsymbol{\theta}\rangle - \boldsymbol{\Omega}\big). \tag{21.4.2}$$

The field $\boldsymbol{H}$ came out of the average because it is common to all grains, and $\boldsymbol{\Omega}$ likewise.

Step 3. Introduce the macroscopic symbols $I \equiv n I_p$, $\boldsymbol{\Theta} \equiv \langle\boldsymbol{\theta}\rangle$, $\boldsymbol{M} \equiv n\langle\boldsymbol{m}\rangle$:

$$I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - n\zeta_r \big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{21.4.3}$$

Step 4. Replace $n\zeta_r$ by I/τ_s using the identity (7.5.12):

$$I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s} \big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big), \tag{SM-8}$$

which is Eq. (SM-8). ■

21.5 Physical Interpretation

The equation is a tug of war. The magnetic torque $\boldsymbol{M} \times \boldsymbol{H}$ tries to twist the grains so as to align their moments with the field; the viscous torque tries to make them turn with the liquid. The inertial term on the left says how fast the compromise is reached — and by Eq. (SM-1) the answer is 10^{-11} s, i.e. instantly. Chapter 24 exploits exactly that.

21.6 Connection to the Next Equation

Equations (SM-6)–(SM-8) involve the fields $\boldsymbol{v}$, p, $\boldsymbol{M}$, $\boldsymbol{\Theta}$ and $\boldsymbol{H}$. Two things are still missing: the constraints that $\boldsymbol{v}$ and $\boldsymbol{H}$ must satisfy (incompressibility and magnetostatics), and the explicit formula for $\boldsymbol{M}_0$. These are Eqs. (SM-9) and (SM-10).

Chapter 22

Equation (SM-9): The Constraints

$$\nabla \cdot \boldsymbol{v} = 0, \qquad \nabla \times \boldsymbol{H} = 0, \qquad \nabla \cdot \big(\boldsymbol{H} + 4\pi \boldsymbol{M}\big) = 0. \tag{SM-9}$$

22.1 Context

The paper introduces these three relations with the remark that they indicate "that the fluid is considered to be incompressible and nonconducting". They are not dynamical equations; they are constraints that the fields must satisfy at every instant. The first was derived in Volume I; the other two are magnetostatics and are derived here.

22.2 Symbols

$\boldsymbol{B} = \boldsymbol{H} + 4\pi \boldsymbol{M}$ is the magnetic induction (G) in Gaussian units; $\boldsymbol{E}$ the electric field; $\boldsymbol{D}$ the electric displacement; $\boldsymbol{J}$ the free current density; $c = 3 \times 10^{10}\ \mathrm{cm\,s^{-1}}$ the speed of light. All other symbols are as before.

22.3 Assumptions

Assumption 22.3.1 (Maxwell's equations)**.** In Gaussian units, for a medium at rest,

$$\nabla \cdot \boldsymbol{D} = 4\pi \rho_{\mathrm{free}}, \qquad \nabla \times \boldsymbol{H} = \frac{4\pi}{c}\boldsymbol{J} + \frac{1}{c}\frac{\partial \boldsymbol{D}}{\partial t}, \tag{22.3.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \qquad \nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t}, \tag{22.3.2}$$

with $\boldsymbol{B} = \boldsymbol{H} + 4\pi \boldsymbol{M}$.

These are empirical laws — Gauss's law, the Ampère–Maxwell law, the absence of magnetic monopoles, and Faraday's law — distilled from two centuries of experiment. They cannot be derived from anything more primitive, and we adopt them in the same spirit as Newton's laws in Sec. 4.1. Everything below is derived.

Assumption 22.3.2 (Non-conducting medium)**.** The carrier is water with dissolved counterions; on the time scales of the experiment no significant free current flows in response to the alternating field. Hence $\boldsymbol{J} = 0$.

Assumption 22.3.3 (Quasi-static regime)**.** The displacement current is negligible.

Justification of Assumption 22.3.3, quantitatively. Compare the two terms on the right of the Ampère–Maxwell law. With a field varying at angular frequency ω and a system of linear size L, the displacement-current term is of order $\omega D/c$ and the curl on the left of order H/L. Their ratio is

$$\frac{\omega D L}{c H} \sim \frac{\omega L}{c} = \frac{2\pi L}{\lambda}, \qquad \lambda = \frac{2\pi c}{\omega} = \frac{c}{f}. \tag{22.3.3}$$

For the highest frequency used by Bacri *et al.*, $f = 1480$ Hz,

$$\lambda = \frac{3 \times 10^{10}}{1.48 \times 10^{3}} = 2.0 \times 10^{7} \text{ cm} = 200 \text{ km}, \tag{22.3.4}$$

while the apparatus is $L \sim 10$ cm. The ratio is $2\pi \times 10/2 \times 10^{7} \approx 3 \times 10^{-6}$. The field fills the solenoid as if it were static, adjusting itself instantaneously to the coil current. ✓

22.4 Step-by-Step Derivation

First relation. Conservation of mass with constant density gives $\nabla \cdot \boldsymbol{v} = 0$; this is Eq. (5.1.4), derived in Sec. 5.1. ✓

Second relation. Insert Assumptions 22.3.2 and 22.3.3 into the Ampère–Maxwell law (22.3.1):

$$\nabla \times \boldsymbol{H} = \frac{4\pi}{c} \underbrace{\boldsymbol{J}}_{=0} + \frac{1}{c} \underbrace{\frac{\partial \boldsymbol{D}}{\partial t}}_{\text{negligible}} = 0. \tag{22.4.1}$$

■

A useful consequence, used in Step 3 of Sec. 19.4 and again in Chapter 27: a curl-free field is a gradient,

$$\nabla \times \boldsymbol{H} = 0 \quad \Longrightarrow \quad \boldsymbol{H} = -\nabla \psi, \tag{22.4.2}$$

for some scalar magnetic potential ψ. (Reason: by Stokes' theorem (2.4.6) the circulation of $\boldsymbol{H}$ around every closed loop vanishes, so the line integral between two points is path-independent and defines a potential.)

Third relation. The absence of magnetic monopoles, $\nabla \cdot \boldsymbol{B} = 0$ from (22.3.2), together with the constitutive definition $\boldsymbol{B} = \boldsymbol{H} + 4\pi \boldsymbol{M}$, gives directly

$$\nabla \cdot \big(\boldsymbol{H} + 4\pi \boldsymbol{M} \big) = 0, \qquad \text{equivalently} \qquad \nabla \cdot \boldsymbol{H} = -4\pi \nabla \cdot \boldsymbol{M}. \tag{22.4.3}$$

■

22.5 Physical Interpretation: why $\boldsymbol{H}$ inside is not the applied field

Equation (22.4.3) says that the magnetisation acts as a source of $\boldsymbol{H}$: wherever $\boldsymbol{M}$ has a divergence — in practice, at the surface of the sample, where $\boldsymbol{M}$ drops abruptly to zero — there are effective "magnetic charges" which create an additional field inside the sample. That field opposes $\boldsymbol{M}$ and is called the demagnetising field. It is the reason a magnetised body does not experience the field one applied to it.

Why this does not spoil the experiment. Bacri *et al.* place a capillary of diameter 1 mm and length several centimetres *along* the axis of the solenoid, so that $\boldsymbol{H}$, $\boldsymbol{M}$ and the tube axis are all parallel. Inside such a long cylinder with $\boldsymbol{M}$ uniform and axial,

- $\nabla \cdot \boldsymbol{M} = 0$ in the interior (a uniform vector has zero divergence), so there are no volume charges;
- the only surface charges lie on the two end faces, at a distance $\gtrsim 10^2$ diameters from the measuring region, so their field there is negligible (it falls off as the inverse square of the distance).

Hence $\boldsymbol{H}$ inside the capillary equals the applied solenoid field, and the demagnetising correction — which for a long cylinder magnetised along its axis is formally zero — may be ignored. This is why the papers can treat H as the externally imposed quantity without further comment. Had the field been applied *across* the capillary, the demagnetising factor would have been 2π per unit M and the internal field would have differed appreciably from the applied one.

A second reason $\boldsymbol{H}$ may be treated as given. Suppose we tried to keep the coupling: $\boldsymbol{M}$ depends on $\boldsymbol{H}$ through the Langevin law, and $\boldsymbol{H}$ depends on $\boldsymbol{M}$ through (22.4.3). The problem would be self-consistent and nonlinear. The geometry above cuts that loop, which is a piece of experimental design, not of mathematics.

22.6 Connection to the Next Equation

The system is now closed except for one thing: the explicit formula for the instantaneous equilibrium magnetisation $\boldsymbol{M}_0$ appearing in (SM-7). That is (SM-10).

Chapter 23

Equation (SM-10): The Langevin Magnetisation

$$\boldsymbol{M}_0 = nmL(\xi)\frac{\boldsymbol{H}}{H}, \qquad \xi = \frac{mH(t)}{kT}, \qquad L(\xi) = \coth\xi - \xi^{-1}. \tag{SM-10}$$

23.1 Context

The last member of the system (SM-6)–(SM-10). It supplies the equilibrium towards which (SM-7) relaxes, and it is the only place where the temperature enters explicitly. Bacri *et al.* write the same formula beneath their Eq. (B-4) as $\boldsymbol{M}_0(t) = \varphi M_s L(\xi)\hat{\boldsymbol{h}}$; the two agree because $nm = nM_sV = \varphi M_s$.

23.2 Symbols

Symbol	Meaning	Type	Units
$\boldsymbol{M}_0$	instantaneous-equilibrium magnetisation	vector field	G
n	number of grains per unit volume	scalar	cm^{-3}
$m = M_sV$	magnetic moment of one grain	scalar	$\mathrm{erg\,G}^{-1}$
$L(\xi)$	Langevin function	function	—
$\xi = mH(t)/kT$	Langevin parameter, evaluated at the *instantaneous* field	scalar	—
$\hat{\boldsymbol{h}} = \boldsymbol{H}/H$	unit vector along the field	vector	—

23.3 Assumptions

1. Thermal equilibrium of the orientations *at the instantaneous field* — i.e. $\boldsymbol{M}_0$ is defined as the magnetisation the sample would have if τ_B were zero. It is a mathematical construct, not the actual magnetisation.

2. No interparticle magnetic interaction.

3. Monodisperse, rigid dipoles.

23.4 Derivation

Carried out in full in Chapter 10: the Boltzmann distribution (9.5.7) gives $\langle\cos\theta\rangle = L(\xi)$ by Eq. (10.2.16), and multiplication by nm gives (10.4.1), which *is* (SM-10). Nothing is repeated.

23.5 Physical Interpretation and a Warning About the Word "Equilibrium"

Shliomis & Morozov insert a paragraph of clarification at exactly this point, and it is worth restating because it is the most common source of confusion in the subject:

> $\boldsymbol{M}_0[\boldsymbol{H}(t)]$ is *not* the magnetisation. It is the value the magnetisation would take if the field were changing slowly enough for the orientations to keep up. The real $\boldsymbol{M}(t)$ differs from it precisely because $\tau_B \neq 0$ in (SM-7).

The difference $\boldsymbol{M} - \boldsymbol{M}_0$ is what drives everything: it is proportional to the transverse magnetisation, hence to the torque $\boldsymbol{M}\times\boldsymbol{H}$, hence to the slip $\boldsymbol{\Omega}-\boldsymbol{\Theta}$, hence to $\Delta\eta$. Chapter 15 computed that chain for a stationary field; Volumes IV and V will compute it for an alternating one.

Numerical calibration of ξ for the experimental fluid. With $M_s = 400$ G and $d = 10$ nm ($V = 5.24\times10^{-19}$ cm^3),

$$m = M_s V = 400\times 5.24\times10^{-19} = 2.10\times10^{-16}\ \text{erg}\,\text{G}^{-1}, \tag{23.5.1}$$

$$\xi = \frac{mH}{kT} = \frac{2.10\times10^{-16}}{4.14\times10^{-14}}H = 5.1\times10^{-3}\,H\ \text{(Oe)}. \tag{23.5.2}$$

Bacri *et al.* quote $\xi_0 = M_sVH/kT = 5.25\times10^{-3}H(\text{Oe})$. Working backwards from their number, $V = 5.43\times10^{-19}$ cm^3 and $d = 10.1$ nm — confirming both their stated grain size and our arithmetic. ✓ Note the practical consequence: their maximum field of 2000 Oe corresponds to $\xi \approx 10.5$, well into the nonlinear region of $L(\xi)$, which is why the weak-field theory of Sec. II of Shliomis & Morozov cannot describe the experiment quantitatively and Sec. III is needed.

23.6 Connection to the Next Equation

The system (SM-6)–(SM-10) is now complete and closed: five relations for $\boldsymbol{v}, p, \boldsymbol{M}, \boldsymbol{\Theta}, \boldsymbol{H}$. The paper's next move is to simplify it by eliminating $\boldsymbol{\Theta}$, which is possible because of the smallness of τ_s. That is (SM-11).

Part VIII

Elimination of the Spin, and the Alternating Field

Chapter 24

Equation (SM-11): The Inertialess Spin

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \big(\tau_s / I\big)\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-11}$$

24.1 Context

This is where the licence bought in Volume I is spent. Equation (SM-1) established that $\tau_s \sim 10^{-11}$ s; the paper now argues that the inertial term of (SM-8) is therefore negligible, converting a differential equation for $\boldsymbol{\Theta}$ into an algebraic one. From this point on, $\boldsymbol{\Theta}$ is no longer an independent unknown: it is whatever the torques say it is.

24.2 Symbols

All previously defined. Note $\tau_s / I = 1/n\zeta_r = 1/6\eta\varphi$ by Eq. (7.5.12), so (SM-11) may equivalently be written

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M} \times \boldsymbol{H}\big), \tag{24.2.1}$$

which is the form used in Volume II [Eq. (11.4.2) averaged, and Sec. 13.3] and the form in which Bacri *et al.* write it as their Eq. (B-1).

24.3 Assumptions

The single assumption is that the inertial term of (SM-8) is negligible:

$$\left| I \frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} \right| \ll \left| \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big) \right|. \tag{24.3.1}$$

Justification, quantitatively. Estimate each side. The left-hand side is of order $I\,\Theta_{\mathrm{typ}}/t_{\mathrm{typ}}$, where t_{typ} is the shortest time on which $\boldsymbol{\Theta}$ actually varies. In these experiments that is the field period, $t_{\mathrm{typ}} = 1/\omega$. The right-hand side is of order $I\,\Theta_{\mathrm{typ}}/\tau_s$. The ratio is therefore

$$\frac{|I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t|}{|I(\boldsymbol{\Theta} - \boldsymbol{\Omega})/\tau_s|} \sim \omega\tau_s. \tag{24.3.2}$$

At the highest experimental frequency, $f = 1480$ Hz,

$$\omega\tau_s = 2\pi \times 1480 \times 10^{-11} = 9.3 \times 10^{-8}. \tag{24.3.3}$$

The inertial term is a hundred-millionth of the friction term. This is precisely the paper's statement "$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$, so Eq. (8) has the form (11)". ✓

24.4 Step-by-Step Derivation

Step 1. Write (SM-8) again:

$$I\frac{\mathrm{d}\boldsymbol{\Theta}}{\mathrm{d}t} = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{24.4.1}$$

Step 2. Set the left-hand side to zero by (24.3.1):

$$0 = \boldsymbol{M} \times \boldsymbol{H} - \frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big). \tag{24.4.2}$$

Step 3. Solve for $\boldsymbol{\Theta}$. Add $\frac{I}{\tau_s}(\boldsymbol{\Theta} - \boldsymbol{\Omega})$ to both sides:

$$\frac{I}{\tau_s}\big(\boldsymbol{\Theta} - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}. \tag{24.4.3}$$

Multiply both sides by τ_s/I:

$$\boldsymbol{\Theta} - \boldsymbol{\Omega} = \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big), \tag{24.4.4}$$

and add $\boldsymbol{\Omega}$:

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-11}$$

■

Equation (24.4.3), written with $I/\tau_s = 6\eta\varphi$, is exactly Eq. (B-1) of Bacri *et al.*,

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

already used in Sec. 14.2.

24.5 Physical Interpretation

The grains have no memory of their own rotation. At every instant the magnetic torque and the viscous torque balance exactly, and the spin is whatever makes that true. Three readings:

- **No magnetic torque $\Rightarrow \boldsymbol{\Theta} = \boldsymbol{\Omega}$**: the grains turn with the liquid, and nothing happens.
- **The slip is proportional to the torque**, with the proportionality constant $1/6\eta\varphi$: more viscous liquid or more particles means less slip for a given torque.
- **The sign of the slip is the sign of the effect.** A stationary field always produces a torque opposing the rotation ($\Theta < \Omega$, $\Delta\eta > 0$), because $\boldsymbol{M}$ lags *behind* $\boldsymbol{H}$ in the direction of rotation. An alternating field can produce the opposite, and that is the subject of the volumes to come.

24.6 Connection to the Next Equation

With $\boldsymbol{\Theta}$ eliminated, both (SM-6) and (SM-7) can be rewritten without it. Doing so gives (SM-12) and (SM-13), the reduced pair the papers actually solve.

Chapter 25

Equation (SM-12): The Reduced Momentum Equation

$$\rho\,\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{SM-12}$$

25.1 Context

Equation (SM-6) with $\boldsymbol{\Theta}$ eliminated by (SM-11). It is also, term for term, Eq. (B-3) of Bacri *et al.* The virtue of this form is that the mechanical unknown $\boldsymbol{\Theta}$ has disappeared entirely: the fluid equation now couples only to the magnetisation.

25.2 Symbols and Assumptions

As in Chapters 19 and 24; nothing new.

25.3 Step-by-Step Derivation

Step 1. Take the last term of (SM-6) and substitute (SM-11) in the form (24.4.3), $\boldsymbol{\Theta}-\boldsymbol{\Omega} = (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$:

$$\frac{I}{2\tau_s}\nabla\times\big(\boldsymbol{\Theta}-\boldsymbol{\Omega}\big) = \frac{I}{2\tau_s}\nabla\times\Big[\frac{\tau_s}{I}\big(\boldsymbol{M}\times\boldsymbol{H}\big)\Big] \tag{25.3.1}$$

$$= \frac{I}{2\tau_s}\cdot\frac{\tau_s}{I}\,\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big) \quad (\tau_s/I \text{ is a constant, so it passes through } \nabla\times) \tag{25.3.2}$$

$$= \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{25.3.3}$$

Step 2. Substitute (25.3.3) into (SM-6):

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{SM-12}$$

■

Note that we arrived at the same expression by an entirely independent route in Sec. 14.3, where $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$ was obtained directly from the requirement of angular-momentum

balance for a body couple. That the two derivations agree is a strong internal check on the whole construction: the couple transmitted by friction and the couple imposed by the field are the same couple.

25.4 Physical Interpretation

The magnetic field enters the momentum balance in exactly two ways:

- through $(\boldsymbol{M} \cdot \nabla)\boldsymbol{H}$, if the field is *non-uniform* — a force;
- through $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$, if the magnetisation is *not parallel* to the field — a couple.

In these papers the first is zero by construction (uniform solenoid field) and the second is everything. It vanishes when $\boldsymbol{M} \parallel \boldsymbol{H}$, i.e. in perfect equilibrium; it is nonzero only because the magnetisation lags.

25.5 Connection to the Next Equation

The same substitution must now be made in the magnetisation equation, giving (SM-13).

Chapter 26

Equation (SM-13): The Reduced Magnetisation Equation

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big) - \frac{\tau_s}{I}\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{SM-13}$$

26.1 Context

Equation (SM-7) with $\boldsymbol{\Theta}$ eliminated. Together with (SM-12) it forms the pair the authors describe as "the equivalent system (10)–(13)", which "allows the calculation of the effective viscosity of the ferrofluid in the stationary and alternating magnetic field". Bacri *et al.* print exactly this equation as their Eq. (B-4), writing $1/6\eta\varphi$ in place of τ_s/I.

26.2 Symbols and Assumptions

As before. The new feature is that the equation is now *nonlinear* in $\boldsymbol{M}$: the last term is cubic in the magnetisation–field product. This nonlinearity is not an artefact — it is the feedback by which the magnetisation changes the particle spin, which in turn changes the magnetisation.

26.3 Step-by-Step Derivation

Step 1. Start from (SM-7) and substitute (SM-11):

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \qquad \boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{26.3.1}$$

Step 2. Expand the convective term, using the distributivity of the cross product over addition:

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \Big[\boldsymbol{\Omega} + \frac{\tau_s}{I}\big(\boldsymbol{M} \times \boldsymbol{H}\big)\Big] \times \boldsymbol{M} \tag{26.3.2}$$

$$= \boldsymbol{\Omega} \times \boldsymbol{M} + \frac{\tau_s}{I}\Big[\big(\boldsymbol{M} \times \boldsymbol{H}\big) \times \boldsymbol{M}\Big]. \tag{26.3.3}$$

Step 3. Reverse the order of the last cross product, using the antisymmetry (1.5.5), $\boldsymbol{A} \times \boldsymbol{B} = -\boldsymbol{B} \times \boldsymbol{A}$:

$$\big(\boldsymbol{M} \times \boldsymbol{H}\big) \times \boldsymbol{M} = -\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{26.3.4}$$

Step 4. Substitute (26.3.4) into (26.3.3) and then into Step 1:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega} \times \boldsymbol{M} - \frac{\tau_s}{I}\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big) - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \tag{SM-13}$$

which is (SM-13) with the terms in the paper's order. ■

26.4 Physical Interpretation of the New Term

Use the triple product (1.7.1) to expand it:

$$\boldsymbol{M} \times \big(\boldsymbol{M} \times \boldsymbol{H}\big) = \boldsymbol{M}\big(\boldsymbol{M} \cdot \boldsymbol{H}\big) - \boldsymbol{H}\big(\boldsymbol{M} \cdot \boldsymbol{M}\big) \tag{26.4.1}$$

$$= -M^2 \left[\boldsymbol{H} - \hat{\boldsymbol{M}}\big(\hat{\boldsymbol{M}} \cdot \boldsymbol{H}\big)\right], \qquad \hat{\boldsymbol{M}} = \boldsymbol{M}/M, \tag{26.4.2}$$

i.e. it is proportional to *minus* the component of $\boldsymbol{H}$ perpendicular to $\boldsymbol{M}$. With the overall minus sign in (SM-13), the term therefore pushes $\boldsymbol{M}$ *towards* $\boldsymbol{H}$. It is a second alignment mechanism, distinct from the Brownian relaxation term: here the alignment is achieved by physically rotating the grains, whereas the relaxation term achieves it by thermal reorientation of the population. The characteristic rate of the mechanical mechanism is

$$\frac{\tau_s}{I} MH = \frac{MH}{6\eta\varphi}, \tag{26.4.3}$$

and the ratio of this rate to the Brownian rate $1/\tau_B$ is exactly the dimensionless combination $\zeta = \tau_B MH/6\eta\varphi = \xi L(\xi)/2$ met in Eq. (15.5.17). The competition between these two alignment mechanisms is what produced the factor $\zeta/(1+\zeta)$ in Eq. (SM-2).

26.5 Connection to the Next Equation

The system (SM-10)–(SM-13) is now the complete, closed problem. What remains is to specify *which* field is applied. That is (SM-14).

Chapter 27

Equation (SM-14): The Alternating Field

$$H_x = H_0 \cos \omega t, \qquad H_y = H_z = 0. \tag{SM-14}$$

27.1 Context

Everything until now has been general. Equation (SM-14) specialises the problem to the case the paper is about: a spatially uniform, *linearly polarised* field oscillating harmonically along a fixed direction. The authors introduce it with the sentence "Thus the problem is to express the viscosity $\Delta\eta$ as a function of amplitude and frequency of the field". From here on, the two control parameters are H_0 (through $\xi_0 = mH_0/kT$) and ω (through $\omega\tau_B$).

27.2 Symbols

Symbol	Meaning	Type	Units
H_0	amplitude of the field, i.e. its maximum value	scalar, constant	Oe
ω	angular frequency, $\omega = 2\pi f$	scalar, constant	s^{-1}
f	ordinary frequency	scalar	Hz
t	time	scalar	s
$\xi_0 = mH_0/kT$	Langevin parameter formed with the *amplitude*	dimensionless	—
$\omega\tau_B$	dimensionless frequency	dimensionless	—

27.3 Assumptions

1. **Spatial uniformity.** H_0 does not depend on position, so the Kelvin force $(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$ vanishes. Realised experimentally by placing the short capillary in the middle of a long solenoid.

2. **Linear polarisation.** The field oscillates along a fixed axis, rather than rotating. This is the essential feature: a linearly polarised field *picks out no sense of rotation*, and the whole mechanism of the effect depends on that fact, as the next volume shows.

3. **Harmonic time dependence.** A pure cosine. Section III of the paper relaxes this and treats a square wave instead, for computational convenience (Eqs. SM-36 onwards), and the two give quantitatively similar answers — a robustness check worth noting.

4. **The field is perpendicular to the vorticity**, $\beta = \pi/2$. The authors state this just before Eq. (SM-3): "below we will assume that $\beta = \pi/2$". It is realised in the experiment because the flow is a cylindrical Poiseuille flow whose vorticity is azimuthal while the field is axial (Sec. 3.5), so $\boldsymbol{H} \perp \boldsymbol{\Omega}$ automatically. It is also the case that maximises the effect, since (SM-2) carries a factor $\sin^2\beta$.

27.4 Mathematical Background: Period Averages

The function $H_0 \cos\omega t$ is periodic with period

$$T_0 = \frac{2\pi}{\omega} = \frac{1}{f}, \tag{27.4.1}$$

its mean over a period is zero,

$$\overline{\cos\omega t} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \cos\omega t\,\mathrm{d}t = \frac{\omega}{2\pi}\left[\frac{\sin\omega t}{\omega}\right]_0^{2\pi/\omega} = \frac{\sin 2\pi - \sin 0}{2\pi} = 0, \tag{27.4.2}$$

and its mean square is one half,

$$\overline{\cos^2\omega t} = \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \cos^2\omega t\,\mathrm{d}t \tag{27.4.3}$$

$$= \frac{\omega}{2\pi}\int_0^{2\pi/\omega} \frac{1+\cos 2\omega t}{2}\,\mathrm{d}t \qquad \text{using } \cos^2 x = \tfrac{1}{2}(1+\cos 2x) \tag{27.4.4}$$

$$= \frac{\omega}{2\pi}\left[\frac{t}{2} + \frac{\sin 2\omega t}{4\omega}\right]_0^{2\pi/\omega} = \frac{\omega}{2\pi}\cdot\frac{\pi}{\omega} = \frac{1}{2}. \tag{27.4.5}$$

Similarly

$$\overline{\sin\omega t\cos\omega t} = \overline{\tfrac{1}{2}\sin 2\omega t} = 0. \tag{27.4.6}$$

These three averages are used repeatedly from Eq. (SM-25) onwards, when the fast oscillation is averaged out of the torque; they are recorded here so that the algebra of Volume IV can proceed without interruption.

27.5 Physical Interpretation, and Why Linear Polarisation Is the Crux

Consider what a linearly polarised field does to a grain in a fluid *at rest*. During the first half-cycle the field points along $+x$ and the moment swings towards it; during the second half-cycle it points along $-x$ and the moment swings back. The grain therefore performs a rotational *oscillation*, not a rotation. Crucially, for every grain that happens to swing clockwise there is another, symmetrically placed, that swings counterclockwise: the field distinguishes the axis x, but not the sense of rotation about z. Averaged over the many grains in a volume element, the mean angular velocity vanishes,

$$\boldsymbol{\Theta} = 0 \qquad \text{in a quiescent fluid.} \tag{27.5.1}$$

Both papers stress this in their abstracts: "one half of the particles rotate clockwise and the other half counterclockwise. Hence the macroscopic angular velocity of the particles equals zero."

Now switch on a shear. The vorticity $\boldsymbol{\Omega}$ breaks the symmetry between the two senses — it is a pseudo-vector that distinguishes clockwise from counterclockwise — and the degeneracy is lifted. The mean particle spin becomes nonzero, and *it need not have the same sign as it would in a stationary field.* At high frequency it turns out to exceed Ω, so that $\boldsymbol{\Omega} - \boldsymbol{\Theta} < 0$ and, by the master formula (14.4.9), $\Delta\eta < 0$.

That, in one paragraph, is the entire physical mechanism of the papers. The next two volumes turn it into algebra.

Remark 27.5.1 (Circular polarisation is different). If instead $\boldsymbol{H} = (H_0 \cos\omega t,\ H_0 \sin\omega t,\ 0)$, the field itself would single out a sense of rotation, and the grains would be driven around even in a fluid at rest — a magnetic stirrer. That situation, mentioned in the opening sentence of Bacri *et al.*, produces a nonzero $\boldsymbol{\Theta}$ without any flow at all. The linearly polarised case is subtler and more interesting: nothing happens *until* there is a shear, and then the shear is amplified.

27.6 Connection to the Next Equation

To solve the system analytically the authors first restrict to weak fields, where the Langevin function may be linearised. That restriction is (SM-15).

Chapter 28

Equation (SM-15): The Linear Magnetisation Law

$$\boldsymbol{M}_0(t) = \chi \boldsymbol{H}(t), \qquad \chi = \frac{nm^2}{3kT}. \tag{SM-15}$$

28.1 Context

Section II of Shliomis & Morozov bears the title "Rotational viscosity under the low-amplitude alternating field". Equation (SM-15) is what "low-amplitude" means: the field is weak enough that the Langevin function may be replaced by its first term. The authors say so explicitly: this assumption "permits to evade the difficulties which are connected with the nonlinearity of the Langevin function (10)". With it, the whole of Sec. II becomes analytically soluble, and its result is the celebrated Eq. (SM-28).

28.2 Symbols

χ is the initial (linear) magnetic susceptibility: dimensionless in Gaussian units, since M and H are both measured in G/Oe. It is a material constant at fixed temperature, proportional to the concentration and to the *square* of the moment.

28.3 Assumptions

Assumption 28.3.1 (Weak field)**.** $mH_0 \ll kT$, i.e. $\xi_0 \ll 1$.

What this means numerically. With the calibration of Sec. 23.5, $\xi_0 = 5.25 \times 10^{-3} H_0(\mathrm{Oe})$, so $\xi_0 \ll 1$ requires

$$H_0 \ll \frac{1}{5.25 \times 10^{-3}} \approx 190 \text{ Oe}. \tag{28.3.1}$$

The experiments of Bacri *et al.* run to 2000 Oe, i.e. $\xi_0 \approx 10$. *Section II of the theory therefore does not apply quantitatively to the experiment.* This is not a criticism of either paper — it is exactly why Sec. III of Shliomis & Morozov exists, and why Bacri *et al.* compare their data with the finite-amplitude results of Figs. 4 and 5 rather than with Eq. (SM-28). The value of Sec. II is that it is exactly soluble and reveals the mechanism.

28.4 Derivation

Already obtained as Eq. (10.4.2). Repeating only the logic, not the algebra: the small-ξ series (10.3.8) gives $L(\xi) \simeq \xi/3$, so

$$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}} \simeq nm\frac{\xi}{3}\hat{\boldsymbol{h}} = \frac{nm}{3}\cdot\frac{mH}{kT}\hat{\boldsymbol{h}} = \frac{nm^2}{3kT}\boldsymbol{H}, \tag{28.4.1}$$

where $H\hat{\boldsymbol{h}} = \boldsymbol{H}$ was used in the last step. Hence $\chi = nm^2/3kT$. ■

Accuracy. The next term of the series (10.3.8) is $-\xi^3/45$, so the relative error of the linearisation is

$$\frac{\xi^3/45}{\xi/3} = \frac{\xi^2}{15}. \tag{28.4.2}$$

At $\xi = 0.5$ this is 1.7%; at $\xi = 1$ it is 6.7%; at $\xi = 3$ it is 60% and the expansion has failed. The linear theory is thus quantitatively reliable up to about $\xi \simeq 1$, i.e. $H_0 \simeq 200$ Oe for this fluid.

28.5 Physical Interpretation

Why $\chi \propto m^2$. Two factors of the moment appear for the same reason two factors of the field appeared in (SM-3): one because a large moment couples strongly to the field and is therefore well aligned, and one because a large moment contributes strongly to the magnetisation once aligned. The consequence is dramatic in practice — since $m = M_s V \propto d^3$, we have $\chi \propto nd^6$, so a small population of large grains dominates the susceptibility of a polydisperse ferrofluid. It is the same d^6 sensitivity that made the measured τ_B of Sec. 17.5 so much larger than the nominal one.

Why $\chi \propto 1/T$. This is Curie's law, and it says that the alignment achieved by a given field is inversely proportional to the strength of the thermal disorder opposing it.

The estimate for the experimental fluid. With $n = \varphi/V = 0.20/5.24\times10^{-19} = 3.8\times10^{17}$ cm^{-3} and $m = 2.10\times10^{-16}$ erg G^{-1}:

$$\chi = \frac{nm^2}{3kT} = \frac{3.8\times10^{17}\times\left(2.10\times10^{-16}\right)^2}{3\times4.14\times10^{-14}} = \frac{1.68\times10^{-14}}{1.24\times10^{-13}} = 0.14. \tag{28.5.1}$$

A susceptibility of order 0.1 confirms, incidentally, the claim of Sec. 22.6 that demagnetising corrections are modest even before the favourable geometry is taken into account: the internal field differs from the applied one by a factor $1/(1+4\pi\chi N)$ with N the demagnetising factor, and $N = 0$ for a long cylinder magnetised axially.

28.6 Connection to the Next Equation

Everything is now in place: a closed system (SM-10)–(SM-13), a specified field (SM-14), and a linear constitutive law (SM-15). The paper proceeds to solve it, and does so twice — first by decomposing the linearly polarised field into two counter-rotating circular ones, which makes

the mechanism visible, and then by a shorter but more formal direct route. The first solution begins at Eq. (SM-16),

$$\boldsymbol{H}_{\pm} = \big(H_0 \cos\omega t,\ \pm H_0 \sin\omega t,\ 0\big), \qquad \boldsymbol{H} = \tfrac{1}{2}\big(\boldsymbol{H}_{+} + \boldsymbol{H}_{-}\big),$$

and that is where Volume IV begins.

Chapter 29

Status and Road Map

29.1 What has been established in Volume III

Equation	Statement	Where derived
(19.4.1)	$\nabla(\boldsymbol{A}\cdot\boldsymbol{B}) = (\boldsymbol{A}\cdot\nabla)\boldsymbol{B} + \boldsymbol{A}\times(\nabla\times\boldsymbol{B})$, $\boldsymbol{A}$ constant	Sec. 19.4
(19.4.6)	Kelvin force $\boldsymbol{f}_{\mathrm{mag}} = (\boldsymbol{M}\cdot\nabla)\boldsymbol{H}$	Sec. 19.4
(SM-6)	fluid motion with spin coupling $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$	Ch. 19
(SM-7)	$\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta}\times\boldsymbol{M} - (\boldsymbol{M}-\boldsymbol{M}_0)/\tau_B$	Ch. 20 (derived Sec. 13.2)
(SM-8)	$I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t = \boldsymbol{M}\times\boldsymbol{H} - (I/\tau_s)(\boldsymbol{\Theta}-\boldsymbol{\Omega})$	Ch. 21
(SM-9)	$\nabla\cdot\boldsymbol{v} = 0$; $\nabla\times\boldsymbol{H} = 0$; $\nabla\cdot(\boldsymbol{H}+4\pi\boldsymbol{M}) = 0$	Ch. 22
(SM-10)	$\boldsymbol{M}_0 = nmL(\xi)\boldsymbol{H}/H$; $\xi = 5.1\times10^{-3}H(\mathrm{Oe})$ for the experimental fluid	Ch. 23 (derived Ch. 10)
(SM-11)/(B-1)	$\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$; justified by $\omega\tau_s \sim 10^{-7}$	Ch. 24
(SM-12)/(B-3)	momentum equation with $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$	Ch. 25
(SM-13)/(B-4)	nonlinear magnetisation equation	Ch. 26
(SM-14)	$\boldsymbol{H} = (H_0\cos\omega t, 0, 0)$; period averages of cos, $\cos^2$, sin cos	Ch. 27
(SM-15)	$\boldsymbol{M}_0 = \chi\boldsymbol{H}$, $\chi = nm^2/3kT = 0.14$; valid for $H_0 \ll 190$ Oe	Ch. 28

29.2 Cumulative progress

Of the 43 equations of Shliomis & Morozov, numbers 1–15 are now complete; of the 10 equations of Bacri *et al.*, numbers 1, 2, 3, 4 and 10 are complete. What remains:

- **Volume IV** — Eqs. (SM-16)–(SM-24): the decomposition of the linearly polarised field into two counter-rotating circular components; the stationary solution for each; the phase lag α; the appearance of the transverse magnetisation M_y when a vortex breaks the degeneracy; and the relation $\tan\alpha = \omega\tau_B$.

- **Volume V** — Eqs. (SM-25)–(SM-34): the averaged magnetic torque, the separation of

fast and slow time scales, and the central result

$$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\,\frac{1-\omega^2\tau_B^2}{\left(1+\omega^2\tau_B^2\right)^2},$$

whose sign change at $\omega\tau_B = 1$ *is* the negative-viscosity effect; together with the second variant of the solution and the formula $\boldsymbol{\Theta} = \boldsymbol{\Omega}\big[1 - \frac{\xi^2}{6}\frac{1-\xi^2\tau_B^2}{(1+\omega^2\tau_B^2)^2}\big]$.

- **Volume VI** — Eqs. (SM-35)–(SM-43): arbitrary field amplitude; the functions $F(t)$, $G(t)$, $\gamma(t)$; the rectangular-modulation solution; and the numerical results of Figs. 3–5.
- **Volume VII** — Eqs. (B-5)–(B-9): the effective-field method of Martsenyuk, Raikher and Shliomis, derived from the Smoluchowski equation (11.6.1); the function $g(\xi_0, \omega\tau_B)$; and the comparison with the experimental isolines.

End of Volume III. Volume IV begins with Eq. (SM-16).